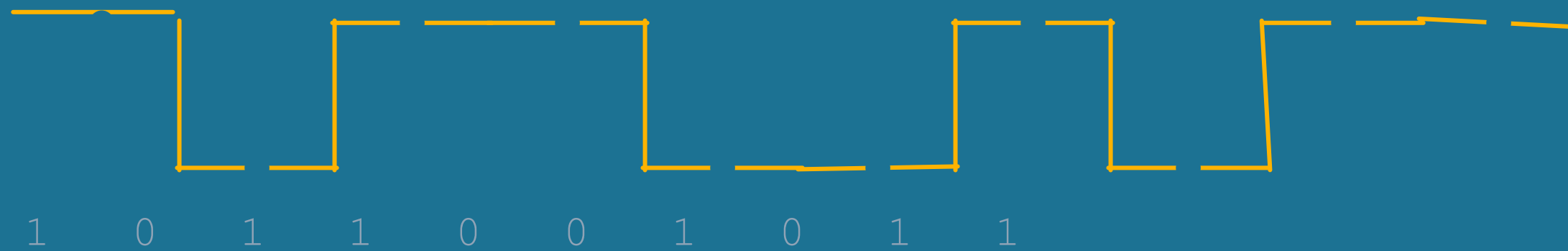


Optical Camera Communication

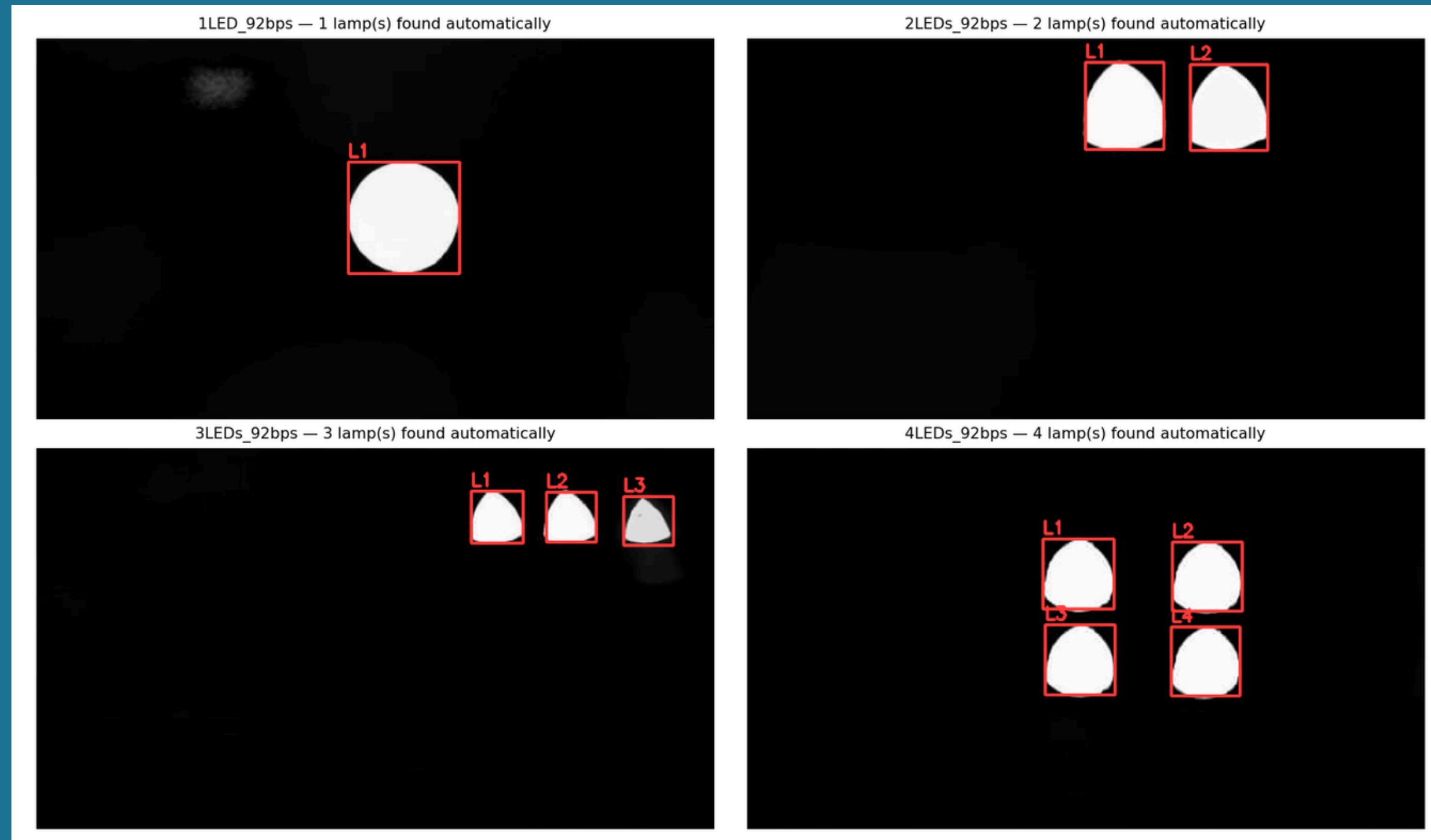
Video Message Decoding



video → bits → message

1 Transmitter ROI — found automatically

Every lamp located from the data alone. No hand-drawn boxes, no lamp count supplied.



How it works

- Group the changing pixels by how their brightness behaves over time.
- Pixels of one lamp blink together. Pixels of different lamps do not.
- This splits touching lamps, which a shape-based method cannot do.
- Halo and glare rejected by fill ratio and relative size.

4 / 4

videos: correct lamp count

0

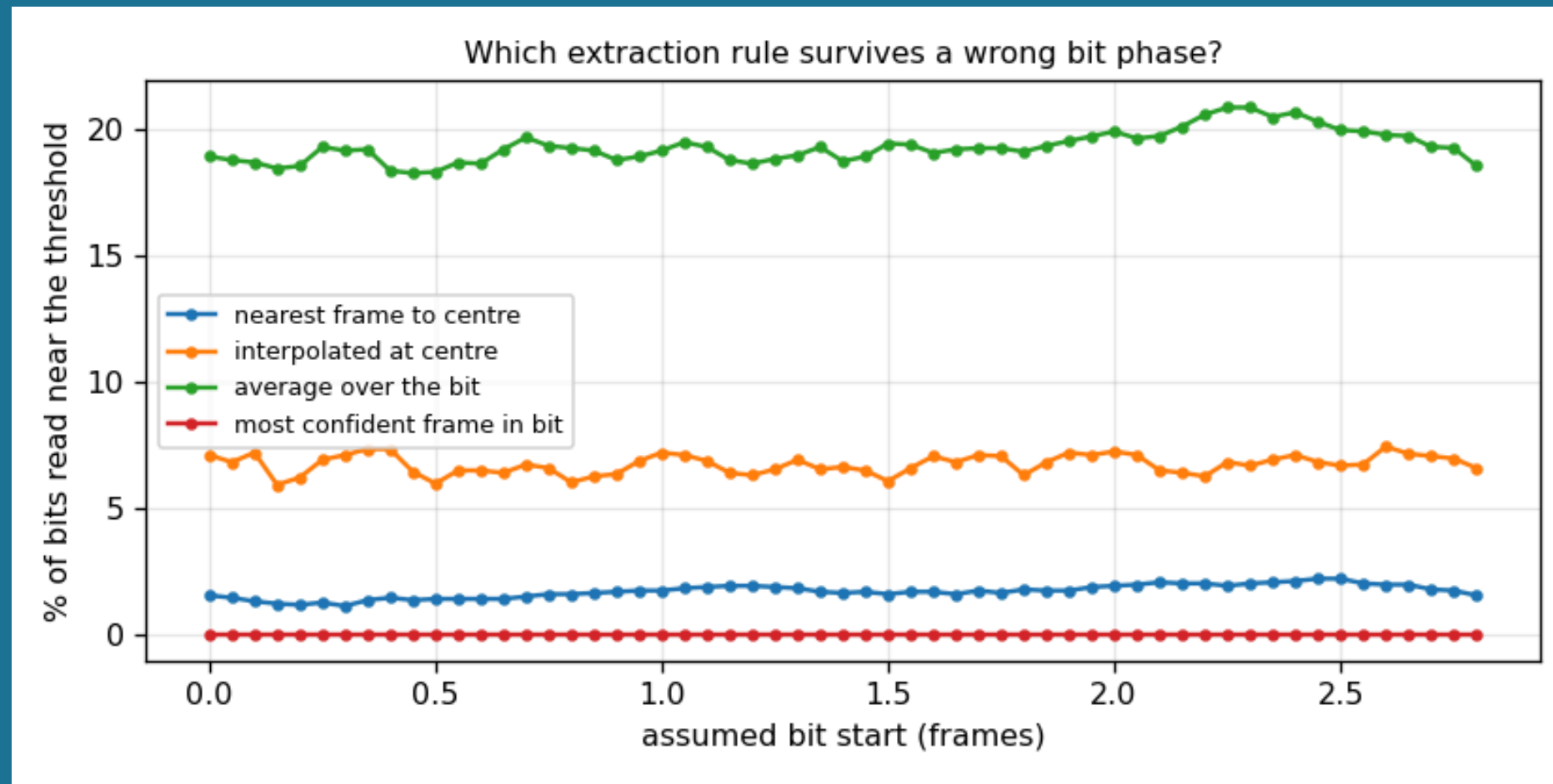
hand-drawn boxes needed

Lamp count was never given to the detector — it is inferred, and it matches the filename in all four videos.

2

Signal extraction — the rule matters more than the timing

2.83 frames per bit means about one frame in three straddles a bit boundary. How we sample decides everything.



Four rules, tested across every bit phase

most confident frame in the bit

0.00 %

nearest frame to the bit centre

1.13 %

interpolated at the bit centre

5.94 %

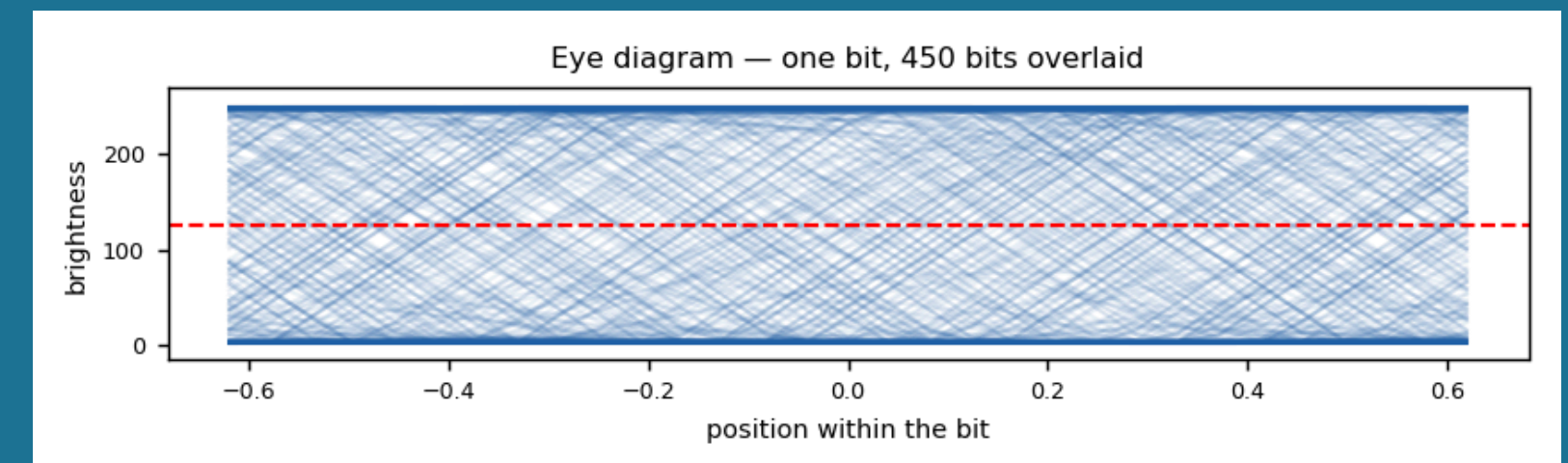
averaged across the whole bit

18.28 %

The eye is NOT cleanly open. Because 2.83 is not a whole number, transitions land at every position inside the bit — so no fixed sampling point is always safe.

But at least one frame per bit always is safe. Picking it scored 0.00% at every phase tested.

Percentage of bits read too close to the threshold to trust. Lower is better.



3 Preprocessing frozen — and what we ask you to approve

Locked today. Any later change needs a log entry and a re-run of everything above it.

Frozen preprocessing

1

Trim

On sustained activity, not the first bright frame

2

Locate lamps

Behaviour clustering; one ROI per lamp

3

Brightness

Mean of the brightest half of each ROI

4

Detrend / normalise

OFF by default — ambient varies 0.1% of contrast

5

Smoothing

OFF — it would erase the edges we read

6

Sample

Most confident frame inside each bit

Synchronisation and evaluation

- Bit rate is known: $260 \text{ fps} \div 92 \text{ bps} = 2.826 \text{ frames per bit}$.
- Phase found by eye-opening search, re-estimated per block over long clips.
- Report the tolerance range, not one accuracy number.
- Tune on the first 30% of each video, report on the last 70%.
- Per-lamp bitstreams saved before any combining.

Blocking — we need a decision

No ground truth was supplied. BER and message correctness cannot be computed on any video, and the ML branch cannot run at all.
The recovered bits are not plain text and not Manchester coded — the framing format is unknown.